

Oh, Wei Yuan

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Languages: English (Business, Native), Chinese (Business, Native)

Availability: July 2026, Internship or Fulltime.

SUMMARY

- MSc Financial Engineering candidate and BSc (Hons, Distinction) in Mathematical Sciences & Economics with focus on quantitative research, statistical modelling and systematic trading.
- Proficient in Python and R for statistical modelling, simulations and data processing, with experience applying these tools to market and risk-related problems.
- Builder-researcher focused on disciplined alpha research: hypothesis → data pipeline → signal construction → leakage-aware validation → backtest diagnostics → live/post-trade review.

SKILLS

- Programming: Python (NumPy, Pandas, SciPy, Matplotlib), C++, R, Excel
- Quantitative & Financial Modelling: statistical modelling, time-series analysis, regularized regression, signal validation, regime detection, Monte Carlo simulation.
- Mathematics & Statistics: Statistical Analysis, Bayesian Inference, Time-series modelling, Simulation, Numerical Methods.

EDUCATION

Master of Science - Financial Engineering Jul 2025 – Jul 2026

Nanyang Technological University, Singapore

- Certificate in Computational Finance (conferred by Carnegie Mellon University)

Bachelor of Science (Hons, Distinction: GPA 4.23/5.00) - Mathematical Science & Economics Aug 2021 – Jul 2025

Nanyang Technological University, Singapore

- Specialisation in Applied Mathematics
- Honours Thesis: Bayesian Inverse Problems for PDEs with Discontinuous Coefficients: An Adaptive Finite Element Approach (A).
- Relevant Coursework: Simulation Techniques in Finance; Financial Econometrics.

PROFESSIONAL EXPERIENCE

NTU Quantitative Asset Management Club, Nanyang Technological University, Singapore Aug 2021 – May 2025

Trading & Research Officer (Trading Team)

- Researched and backtested systematic trading signals in Python, covering feature engineering, model fitting, validation, transaction cost modelling, and performance attribution.
- Applied regularized regression methods including LASSO and Ridge to reduce overfitting and improve out-of-sample robustness.
- Designed and evaluated strategies with explicit transaction costs, risk limits, and disciplined post-mortems.

Advantest Corporation, Singapore

May 2023 – Aug 2023

Corporate Innovation Intern

- Transformed unstructured operational data into structured datasets and dashboards, using Python, SQL and Excel/Power BI to support management reporting, reducing reporting delays by ~24 hours over 7 business regions.
- Automated recurring reporting by generalizing data preparation logic in reusable scripts, lowering manual workload and improving reproducibility of analytics.

SELECTED QUANT SYSTEMS PROJECTS

Systematic Crypto Trading Bot — SG x HK Roostoo Crypto Quant Competition

Grand Prize Winner, Overall Research | Backed by IMC Trading, Optiver, Cubist Systematic Strategies & Gondor Capital

- Built and deployed a 24/7 automated crypto trading bot for a live trading competition, reaching the Singapore finals and winning the overall research grand prize.
- Developed a medium-frequency systematic strategy designed to capture short-term impulse and momentum effects across volatile crypto assets.
- Built an EWMA momentum ranking signal using 6h/24h log-return features, dynamic spread filtering, and entry gates.
- Tested Ridge, XGBoost, LightGBM, and LASSO variants under walk-forward validation; retained the simpler EWMA signal where ML did not improve the full pipeline.
- Built a PCA-HMM regime layer using 43-coin returns → 4 principal components → 3-state HMM, mapping regimes to exposure.
- Evaluated signal quality using IC, Spearman/rank IC, Top-K analysis, live-vs-backtest diagnostics, and leakage-aware forward-return guards.

Kalman Filter Pairs Trading — Dynamic Hedge Ratio

- Built intraday pairs pipeline (tick → 5-min VWAP), aligned pair series, and implemented a state-space Kalman filter to estimate time-varying hedge ratio.
- Traded z-scored Kalman spread via a backtest engine (positions/cash/equity); validated with ADF/CADF/Johansen and ran window/threshold robustness sweeps with performance diagnostics.

Final Year Project – Assoc Prof Hoang Viet Ha

Aug 2024 – Jun 2025

Adaptive Finite Element Inference (Thesis)

- Developed an adaptive finite-element solver and Bayesian inference pipeline in Python for PDEs with discontinuous coefficients, focusing on numerical stability and scalability.
- Emphasized numerical stability, sensitivity to assumptions, and clear documentation of methodology and limitations.
- Optimized performance via vectorization and memory-efficient structures to scale simulation workloads.